systems. He also added that by the end of this year, the feature-sets of traditional hardware, converged and hyper-converged, will align and it will be difficult to even see the differences.

How HCI differs from traditional virtualisation, converged infrastructure and cloud computing

In the beginning, there was just IT infrastructure and no virtualisation. Enterprises spent a lot on buying the best IT components for each function, and spent even more on people to implement and manage these silos of technology.

Then came virtualisation, which greatly optimised the way resources were shared and used without wastage by multiple departments or functions.

In the last decade, we have seen converged infrastructure (CI) emerging as a popular option. In CI, the server, storage and networking hardware are delivered as a single rack, and sold as a single product, after being pre-tested and validated by the vendor. The system may include software to easily manage hardware components and virtual machines (VMs), or support third-party hypervisors like VMware vSphere, Microsoft Hyper-V and KVM. As far as the customer is concerned, CI is convenient because there is just a single point of contact and a single bill. Chris Evans, a UK-based consultant in virtualisation and cloud technologies, describes it humorously in one of his blogs, where he says that in CI there is just one throat to choke!

While this single product stack approach is ideal for most enterprises, there are some who want to do it themselves. This could be because they need heavy customisation or have some healthy legacy infrastructure that they do not want to discard. Given the required time and resources, one can also build a CI out of reference architectures. Here, the vendor gives you the reference architecture, which is something like an instruction manual that tells you which supplier platforms can work together well and in what ways. So, the vendor tests, validates and certifies the interoperability of products and platforms, giving you more than one option to choose from. Thereafter, you can procure the resources and put them together to work as CI.

Some leading players

- Nutanix
- HPE's Simplivity
- Cisco Systems
- NetApp
- Dell EMC and VMware
- Pivot3

Enter HCI. What makes the latest technology 'hyper-converged' is that it brings together a variety of technologies without being vendor-agnostic. Here, multiple off-the-shelf data centre components are brought together below a hypervisor. Along with the hardware, HCI also packages the server and storage virtualisation features. The term 'hyper' also stems from the fact that the hypervisor is more tightly integrated into a hyper-converged infrastructure than a converged one. In a hyper-converged infrastructure, all key data centre functions run as software on the hypervisor.

The physical form of an HCI appliance is the server or node that includes a processor, memory and storage. Each node has individual storage resources, which improves the resilience of the system. A hypervisor or a virtual machine running on the node delivers storage services seamlessly, hiding all complexity from the administrator. Much of the traffic between internal VMs also flows over software-based networking in the hypervisor. HCI systems usually come with features like data de-duplication, compression, data protection, snapshots, wide-area network optimisation, backup and disaster recovery options.

Deploying an HCI appliance is a simple plug-and-play procedure. You just add a node and start using it. There is no complex configuration involved. Scaling is also easy because you can add nodes on the go, as and when required. This makes it a good option for SMEs.

Broadly speaking, CI can be considered a hardware-focused approach, while HCI is largely or entirely software-defined. CI works better in some scenarios, while HCI works for others. Some enterprises might have a mix of traditional, converged and hyper-converged infrastructure. The common thumb rule today is that when organisations don't have a big IT management team and want to invest only in a phased manner, HCI is an ideal option. They can add nodes gradually, depending on their needs and manage the infrastructure with practically no knowledge of server, storage or network management. However, they lose the option of choosing specific storage or server products and have to settle for what their vendor offers. For organisations with more critical applications that require greater control over configuration, a converged system is better. CI offers more choice and control, but users might have to purchase the software themselves, which means more time, money and effort.

Finally, even though HCI is a software-defined architecture that offers the convenience of the cloud to your enterprise infrastructure, it is not a term that can be used synonymously with cloud computing. The two are quite different. The cloud is a service—it provides infrastructure, platform and software as services to those who need it. A cloud could be private, public or hybrid, and herein starts the confusion, as many people mistakenly think that hyper-convergence and the private cloud are one and the same. Hyper-convergence is merely a technology that enables easy deployment of Infrastructure as a Service in your private cloud by addressing many operational issues and improving cost-efficiency. It is an enabler, which prepares your organisation for the cloud era, and not the cloud itself!